Ἵππαρχος ὁ Νικαεύς

Hipparchus of Nicaea and the Reform of Greek Star Catalogues

eul_aid: lwc

Period: *Ἑλληνιστική* Hellenistic
Dialect: *Μικταὶ διάλεκτοι* Mixed Dialects (frg)
Domain: *Ἐπιστήμη* Science
Format: *Πραγματεία* Treatise

Hipparchus of Nicaea was a Greek astronomer, geographer, and mathematician, born in Nicaea in Bithynia and active during the 2nd century BCE. His recorded astronomical observations date from 147 to 127 BCE. He conducted most of his work from Rhodes, and his career marks a pivotal shift in Greek astronomy toward systematic observation and mathematical modeling.

Almost all of Hipparchus's original writings are lost, known primarily through later sources like Ptolemy's Almagest. His only surviving work is a commentary on the Phaenomena of Aratus and Eudoxus. His major contributions, as reported by ancient sources, include a star catalog of roughly 850 stars, studies on the lengths of the year and month, the discovery of the precession of the equinoxes, and works on solar and lunar theory. He also wrote a critique, Against the Geography of Eratosthenes, and produced foundational works in trigonometry, including the first known trigonometric table.

Hipparchus is considered the greatest astronomical observer of antiquity and the founder of trigonometry. His meticulous observations and mathematical models, such as the use of epicycles, provided the empirical foundation for Ptolemaic astronomy. His discovery of precession was a monumental achievement, and his critical methodology influenced later science.

Works (1)

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